

TECHNICAL BULLETIN

DIXIE CHEMICAL

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n-Octadecane (CAS# 593-45-3)

Typical Properties

Properties	Value	Test Method
Molecular Weight	254	NA
Appearance (Molten)	Clear & Colorless Liquid	D-653D (Visual)
Melting Point, °C	26 - 28	D-653C
n-Octadecane, %	> 98	D-653B
Heat of Fusion, J/g	> 210	D-653C
Density at 40°C, g/mL	0.77	D-102
Flash Point	166	ASTM D-93

Applications

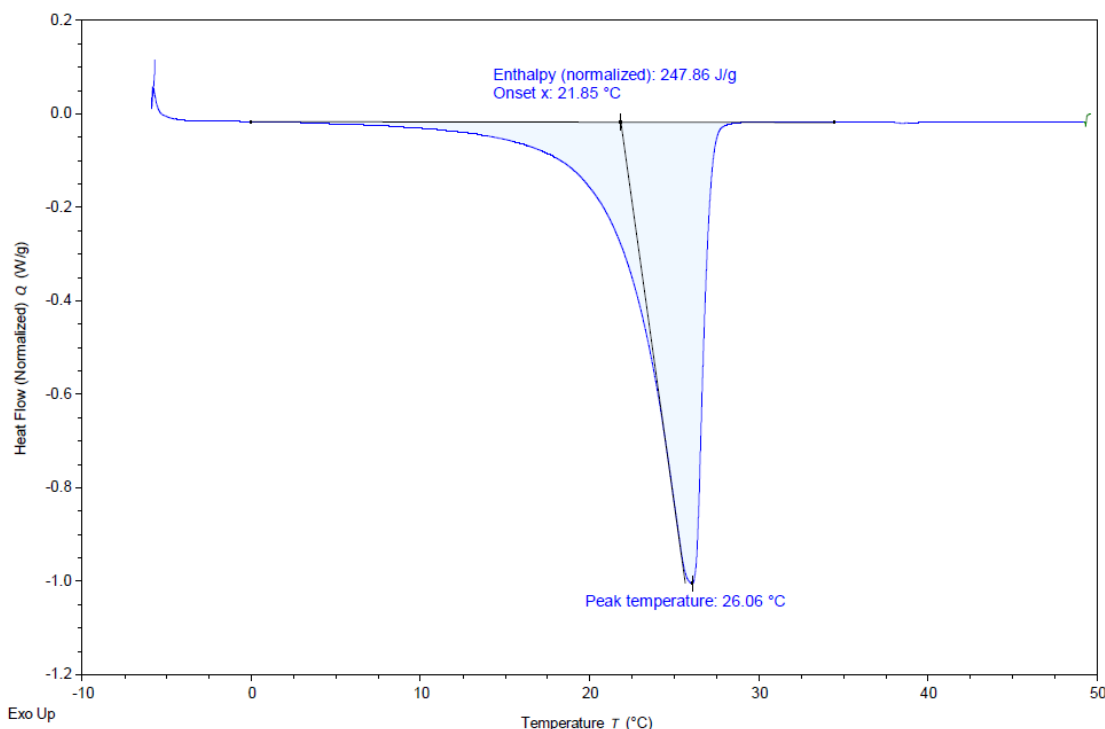
n-Octadecane is a linear non-polar inert hydrocarbon with 18 carbons. It is widely used as an alternative solvent and lubricant for paints, inks, coatings, adhesives, transformer oil, and metalworking/industrial cleaning/drilling fluids. More extensively, it is utilized for latent heat storage due to its sharp melting point as shown in Figure 1. This is a phase change material (PCM) that undergoes solid-liquid phase transitions within a desirable temperature. To enhance the renewable and sustainable energy technologies, we've seen n-Octadecane plays a critical role in efficiency and viability of energy storage, such applications as in construction, solar energy storage, electronic cooling, and medical therapy; shape-stabilized/supported PCM for bedding and functional textile etc.

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Figure 1. DSC scan of a typical n-Octadecane



Handling & Storage

n-Octadecane is very stable under normal conditions. It should be stored away from moisture, high temperatures, ignition sources, or oxidizing materials.

Health Hazards

Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, immediately flush the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

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The minimum personal protective equipment when handling these materials typically includes protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

n-Octadecane is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie Chemical Company representative for details.